

Homework 2 (Due: 4/10)

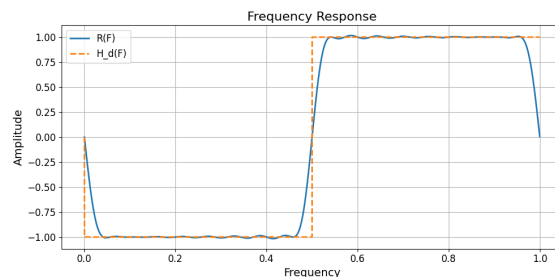
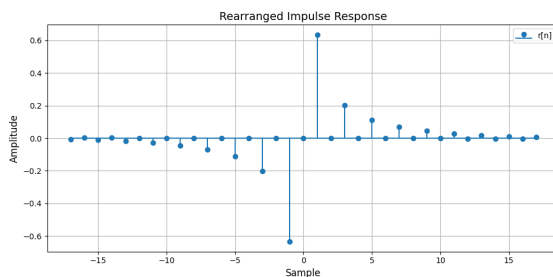
M11202169

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- (1) Write a Matlab or Python code that uses the frequency sampling method to design a $(2k+1)$ -point discrete Hilbert transform filter (k is an input parameter and can be any integer). (25 scores)

The transition band is assigned to reduce the error (unnecessary to optimize). (i) The impulse response and (ii) the imaginary part of the frequency response (DTFT of $r[n]$, see pages 113 and 114) of the designed filter should be shown in the homework. The code should be handed out by NTU Cool.

$k=17$ 時



- (2) Estimate the length of the digital filter if both the passband ripple and the stopband ripple are smaller than 0.01, the sampling interval $\Delta_t = 0.00005$, and the transition band is from 5000Hz to 6000Hz. (10 scores)

$$N = \frac{2}{3} \times \frac{1}{\Delta F} \times \log_{10} \left(\frac{1}{10 \delta_1 \delta_2} \right) \Rightarrow N = \frac{2}{3} \times \frac{1}{0.05} \times \log_{10} \left(\frac{1}{10 \times 0.01 \times 0.01} \right)$$

$$\Delta F = \frac{f_1 - f_2}{f_s} = (f_1 - f_2) \cdot \Delta t \quad \delta_1, \delta_2 \text{ 都代 } 0.01$$

$$= (6000 - 5000) \times 0.00005$$

$$= 0.05$$

$$= 40 *$$

- (3) Why it is improper to use the method of $y[n] = \text{IDFT}(\text{DFT}(x[n])H[m])$ for FIR filter design? (5 scores)

第四種方式：直接把 input 做 DFT, 再設定 $H[m]$ 的 passband、stopband, 最後再做 IDFT

不適合的原因：large computation loading

$\Rightarrow \Theta(N \log N)$ 計算量太大了！

- (4) Derive the way to use the algorithm on page 58-61 to implement an odd symmetric filter with even length (i.e., type 4 on page 90). (10 scores)

• Type 4: $R(F) = \sum_{n=1}^{k+1/2} s[n] \sin(2\pi(n-1/2)F)$ $\begin{matrix} * \\ \alpha = 2\pi nF, \beta = \pi F \end{matrix}$ $\sin(\alpha+\beta) - \sin(\alpha-\beta) = 2\sin\beta \cos\alpha$

$$\sin(2\pi(n+1/2)F) - \sin(2\pi(n-1/2)F) = 2\sin(\pi F) \cos(2\pi nF)$$

判斷 $R(F)$ 能被改寫成: $R(F) = \sin(\pi F) \sum_{n=0}^k S_1[n] \cos(2\pi nF)$

求 $S_1[n]$ 和 $S[n]$ 之間的關係

$$R(F) = \frac{1}{2} \sum_{n=0}^k S_1[n] \sin(2\pi(n+1/2)F) - \frac{1}{2} \sum_{n=0}^k S_1[n] \sin(2\pi(n-1/2)F)$$

$\downarrow n_{old} = n_{new} - 1$

$$= \frac{1}{2} \sum_{n=1}^{k+1} S_1[n-1] \sin(2\pi(n-1/2)F) - \frac{1}{2} \sum_{n=0}^k S_1[n] \sin(2\pi(n-1/2)F)$$

$$= \underbrace{-\frac{1}{2} S_1[0] \sin(-\pi F)}_{n=0 \text{ 時}} + \frac{1}{2} \sum_{n=1}^k (S_1[n-1] - S_1[n]) \sin(2\pi(n-1/2)F) + \underbrace{\frac{1}{2} S_1[k] \sin(2\pi(k+1/2)F)}_{n=k+1 \text{ 時}}$$

$$\left(\text{令 } k+1/2 = k \right) = (S_1[0] - \frac{1}{2} S_1[1]) \sin(\pi F) + \sum_{n=2}^{k-1/2} \frac{1}{2} (S_1[n-1] - S_1[n]) \sin(2\pi(n-1/2)F) + \frac{1}{2} S_1[k-1/2] \sin(2\pi kF)$$

比較係數得:
$$\begin{cases} S[1] = S_1[0] - \frac{1}{2} S_1[1] \\ S[n] = \frac{1}{2} (S_1[n-1] - S_1[n]), \quad n = 2, 3, \dots, k-1/2 \\ S[k+1/2] = \frac{1}{2} S_1[k-1/2] \end{cases}$$

$$e_{rr}(F) = [R(F) - H_d(F)] W(F)$$

$$= \left[\sin(\pi F) \sum_{n=0}^{k-1/2} S_1[n] \cos(2\pi nF) - H_d(F) \right] W(F)$$

$$= \left[\sum_{n=0}^{k-1/2} S_1[n] \cos(2\pi nF) - \cos(\pi F) H_d(F) \right] \sin(\pi F) W(F)$$

⇒ 將 page 58-61 的方法中, $H_d(F)$ 換成 $\cos(\pi F) H_d(F)$

$W(F)$ 換成 $\sin(\pi F) H_d(F)$

k 換成 $k-1/2 = \frac{N}{2} - 1$

- (5) Suppose that $x[n] = 1 + \sin(n)$. (a) What is the Hilbert transform of $x[n]$?
 (b) What is the analytic function corresponding to $x[n]$? (10 scores)

(a) $x[n] = 1 + \sin(n)$, $H(F) = \begin{cases} -j, & 0 < F < 0.5 \\ j, & -0.5 < F < 0 \\ 0, & \text{otherwise} \end{cases}$

$$X(F) = \delta(F) + \frac{1}{2j} \delta(F - \frac{1}{2\pi}) - \frac{1}{2j} \delta(F + \frac{1}{2\pi})$$

$$= \delta(F) - \frac{j}{2} \delta(F - \frac{1}{2\pi}) + \frac{j}{2} \delta(F + \frac{1}{2\pi})$$

$$X_H(F) = -\frac{1}{2} \delta(F - \frac{1}{2\pi}) - \frac{1}{2} \delta(F + \frac{1}{2\pi})$$

$$x_H[n] = -\cos(n) *$$

(b) $x_a[n] = x[n] + jx_H[n]$

$$= 1 + \sin(n) - j\cos(n)$$

$$= 1 + \frac{e^{jn} - e^{-jn}}{2j} - j \frac{e^{jn} + e^{-jn}}{2}$$

$$= 1 - \frac{j}{2} e^{jn} + \frac{j}{2} e^{-jn} - \frac{j}{2} e^{jn} - \frac{j}{2} e^{-jn}$$

$$= 1 - j e^{jn} *$$

- (6) Among the following filters: (i) the Notch filter (ii) the Hilbert transform, (iii) the matched filter, (iv) the difference, (v) the Kalman filter, (vi) the particle filter, and (vii) the Wiener filter,
 (a) Which filters are suitable for edge detection? (b) Which filters are suitable for prediction? (10 scores)

- (a) (ii) the Hilbert transform (v) the Kalman filter
 (iv) the difference (vi) the particle filter *

- (7) (a) What are the two main advantages of the minimum phase filter? (b) Compared to the equalizer, what are the two main advantages of the cepstrum to deal with the multipath problem? (10 scores)

- (a) ① make the energy concentrating on the region near to $n=0$
 ② make both the forward and the inverse transforms stable *
- ↳ 所有 poles 和 zeros 都在單位圓內

- (b) equalizer 在 multipath 會有兩個問題, cepstrum 則不會有
- ① $H(z)$ 可能 unstable
- ② $H(z)$ is usually a dynamic response *

(8) If the z-transform of $h[n]$ is $H(z) = \frac{1+z^{-1}-1.5z^{-2}+z^{-3}}{1-0.3z^{-1}-0.4z^{-2}}$

(a) Determine the cepstrum of $h[n]$.

(b) Convert the IIR filter into the minimum phase filter.

(20 scores)

$$(a) H(z) = \frac{z \bar{z}^{-1} (1 + \frac{1}{2} \bar{z}) (1 - (0.5 + 0.5j) \bar{z}^{-1}) (1 - (0.5 - 0.5j) \bar{z}^{-1})}{(1 - 0.8 \bar{z}^{-1}) (1 + 0.5 \bar{z}^{-1})} \quad \text{因式分解}$$

$$\hat{H}[n] = \begin{cases} \log 2, & n=0 \\ -\frac{(0.5+0.5j)^n + (0.5-0.5j)^n}{n} + \frac{0.8^n - 0.5^n}{n}, & n>0 \\ \frac{-0.5^{-n}}{n}, & n<0 \end{cases}$$

對照 Page 187

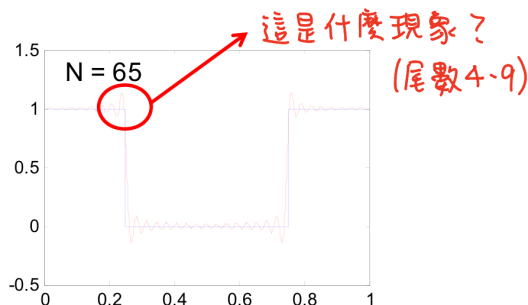
↗

$$\hat{x}[n] = \begin{cases} \log(A), & n=0 \\ -\sum_{k=1}^{m_1} \frac{a_k^n}{n} + \sum_{k=1}^{p_1} \frac{c_k^n}{n}, & n>0 \\ \sum_{k=1}^{m_0} \frac{b_k^{-n}}{n} - \sum_{k=1}^{p_0} \frac{d_k^{-n}}{n}, & n<0 \end{cases}$$

$$\begin{aligned} (b) H(z) &= \frac{z^3 + z^2 - 1.5z + 1}{z^3 - 0.3z^2 - 0.4z} \\ &= \frac{(\cancel{z+2})(z - (0.5+0.5j))(z - (0.5-0.5j))}{z(z-0.8)(z+0.5)} \times (-2) \frac{z+\frac{1}{2}}{\cancel{z+2}} \\ &= \frac{-2(z+\frac{1}{2})(z - (0.5+0.5j))(z - (0.5-0.5j))}{z(z-0.8)(z+0.5)} \quad * \end{aligned}$$

(Extra): Answer the questions according to your student ID number.

(ended with (4, 9), (0, 5), (1, 6), (2, 7))



Ans: Gibbs phenomenon *

⇒ 即使 N 非常大, 邊界附近也會有振盪